

# Vol 14 Chapter 10 — Substrate Complexity Classes

Keeper (author pass — deep math/physics revision)

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## Chapter 10 — Substrate Complexity Classes

### Level 1 — one sentence

Computational complexity classes — P (polynomial time), NP (nondeterministic polynomial), BPP (bounded-error probabilistic), BQP (bounded-error quantum), AC(0) (constant-depth) — organize what can be computed efficiently; BST identifies substrate computation as AC(0)-bounded, and Casey's Curvature Principle proves  $P \neq NP$  via substrate K-type curvature (non-linearizability) obstruction.

### Level 2 — graduate-physicist precision

#### 10.1 Standard complexity hierarchy

- P: polynomial-time deterministic Turing machine
- NP: polynomial-time verifiable; problems with short proofs
- co-NP: complement of NP
- BPP: bounded-error probabilistic polynomial
- BQP: bounded-error quantum polynomial (Shor, Grover)
- PSPACE: polynomial space
- EXPTIME: exponential time

Hierarchy:  $AC^0 \subsetneq NC^1 \subsetneq P \subseteq NP, BPP, BQP \subseteq PSPACE \subseteq EXPTIME$ .

Major open: P vs NP. BPP vs P (likely equal under derandomization). BQP vs BPP (unclear).

#### 10.2 Substrate complexity = AC(0)

BST T421 + T316: substrate computation has depth  $\leq 1-2$  under Casey strict-protocol.

Implication: substrate is AC(0). All BST-derived constants reduce to bounded-depth counting.

Substrate cannot directly compute: - Parity (Hastad 1986) - Majority (Razborov-Smolensky 1987) - Reachability in graphs (Furst-Saxe-Sipser)

Substrate can directly compute: - Addition, multiplication of fixed-precision - BST-primary multiplications and ratios - K-type counting at bounded depth

#### 10.3 Casey's Curvature Principle and $P \neq NP$

Casey: "You can't linearize curvature.  $P \neq NP$  = Gauss-Bonnet for computation."

Mechanism: -  $P$  = problems solvable in linearizable substrate K-type configurations -  $NP$  = problems requiring non-linearizable substrate K-type configurations - Linearization fails for curved K-type kernel (Gauss-Bonnet obstruction) -  $\rightarrow P \neq NP$

Five BST primary integers identified as curvature invariants of substrate K-type space.

#### 10.4 $P \neq NP$ — three independent proved routes (T29 CLOSED)

T1425 (April 23, 2026): AC(0) argument. Triangle-free SAT +  $E[\deg] < 2$  + clustering  $\rightarrow$  algebraic independence  $\rightarrow 2^{\Omega(n)}$  lower bound. Foundation: Toy 1410 (discrete Gauss-Bonnet).

Three independent proved routes: 1. AC(0)/curvature route (T1425) 2. Resolution lower bound (Toy 303) 3. All-P (TCC conditional)

$P \neq NP$  STATUS: T29 CLOSED.

#### 10.5 BQP and substrate

BQP includes Shor's algorithm (factoring in poly-time), Grover (search in  $\sqrt{n}$ ).

BST view: BQP is substrate Bergman-projection-enabled (Vol 14 Ch 5) — quantum measurement IS substrate commitment. Quantum speedups are substrate-natural; classical-only is constrained.

#### 10.6 K-audit anchors

- T29 CLOSED:  $P \neq NP$  via AC(0)
- T1425: triangle-free SAT clustering proof
- T421, T316: depth bounds
- Casey's Curvature Principle

#### Level 3 — 5th-grader accessibility

Complexity classes: P (easy), NP (hard but verifiable), BQP (quantum). Major question: P vs NP. BST: substrate is AC(0) — bounded-depth circuits, can't compute parity.  $P \neq NP$ : Casey's Curvature Principle. "You can't linearize curvature." T29 CLOSED: three independent proved routes including triangle-free SAT clustering (T1425). BQP = quantum advantages from substrate Bergman commitment.

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#### What comes next

Chapter 11 develops information completeness.

#### Where to look this up

- Hastad 1986; Razborov-Smolensky 1987
- BST: T29, T1425, Casey's Curvature Principle